

Artificial Intelligence

Homework 6

Prof. Truong-Son Hy

October 23, 2025

Deadline

October 30, 2025

Note:

- In this assignment, you will have to implement Ant Colony Optimization to solve TSP. You have to submit all of your codes and write a report which details your algorithm with your experimental results.
- You can take a look at my codes during the lecture or given in the github repositories (**actually, it is the solution!**). You can try to understand the codes and re-implement based on your understanding.

Ant Colony Optimization for Travelling Salesman Problem

Based on the description given in the lecture, your task is to implement Ant Colony Optimization to solve the Travelling Salesman Problem. Describe in details how you implement the stochastic path sampling given p_{xy} . Your implementation should be able to return the optimal solution for small N (let's say $N \leq 20$) that can be verified by a brute-force back-tracking algorithm (for $N \leq 10$) and bit-mask dynamic programming (for $N \leq 20$). Report your choice for hyperparameters including $T, \rho, m, Q, \epsilon, \alpha, \beta$; and the best solution you found in each iteration. Please visualize as you did for the previous homework with Evolutionary Algorithm.